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SUBJECT CODE NO: - RNP-8003
FACULTY OF SCIENCE AND TECHNOLOGY
F. E. (NEP) [Group A & B]Mech, Civil, Comp. AI &ML,EEE
Examination January/February 2025
Engineering Chemistry

[Time: 3:00 Hours] **[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B
1) Use of non-programmable calculator is allowed
2) Q.no.1 is compulsory
3) Solve any four questions from Que. nos 2, 3, 4, 5, 6 and 7

SECTION A

Q1 Solve the following questions (any 10) **20**

1) Total hardness of water is 1250 ppm temporary hardness of water is 350 ppm calculate permanent hardness of water.
   a) 900ppm b) 610 ppm
   c) 1250 ppm d) 350 ppm

2) In Carbon-Carbon (C-C) composites, what is the role of the carbon fiber reinforcement?
   a) To increase the material's thermal conductivity
   b) To provide strength, stiffness, and fracture toughness
   c) To enhance the matrix's electrical conductivity
   d) To decrease the weight of the composite material

3) A 100 mL water sample contains 50 mg/L of Ca2+ and 30 mg/L of Mg2+ ions. Calculate the total hardness of the water in mg/L as CaCO3.
   a) 80 mg/L b) 100 mg/L
   c) 120 mg/L d) 150 mg/L

4) Which of the following best describes the storage medium used in Computed Radiography (CR)?
   a) Digital storage devices like hard drives or cloud storage
   b) Photo Stimulable phosphor (PSP) plates that are later scanned
   c) Amorphous selenium detectors
   d) Optical storage films

5) What is the main difference between ultrasonic testing and x-ray testing?
   a) UT uses electromagnetic waves, while x-ray testing uses sound waves
   b) UT requires direct contact with the material, while x-ray testing can be non-contact
   c) UT is used for testing electrical properties, while x-ray testing is used for testing mechanical properties
   d) UT is only used for metals, while x-ray testing is for all materials

6) What is the main difference between Computed Radiography (CR) and Digital Radiography (DR)?
   a) CR uses phosphor plates, while DR uses a direct digital detector
   b) DR uses phosphor plates, while CR uses a direct digital detector
   c) CR uses film, while DR uses a digital sensor
   d) There is no difference between CR and DR

7) What is the primary catholic reaction that occurs during wet corrosion?
   a) $Fe2+ + 2e- \rightarrow Fe$
   b) $02+ 2H2O + 4e- \rightarrow 40H-$
   c) Fe + 2H2O-Fe (OH) 2+2H+
   d) Fe (OH) $2 \rightarrow Fe2O3+H2O$

8) What is the difference between Gross Calorific Value (GCV) and Net Calorific Value (NCV)?
   a) GCV includes the latent heat of water vapour, while NCV does not.
   b) GCV and NCV are the same values.
   c) NCV includes the latent heat of water vapour, while GCV does not.
   d) GCV is always higher than NCV by a fixed amount.

9) In laser cladding, the laser beam is typically used to:
   a) Heat the material to a molten state
   b) Solidify the material instantly
   c) Add a layer of coating material directly to the part surface
   d) Remove any impurities from the base material

10) Which of the following is the primary difference between flash point and fire point?
   a) Flash point is the temperature at which the liquid begins to burn, while fire point is the temperature at which the liquid's vapour ignites.
   b) Flash point is a lower temperature than fire point, and fire point is the temperature at which combustion can be sustained.
   c) Flash point is determined in a closed system, while fire point is determined in an open system.
   d) There is no difference between flash point and fire point.

11) What is the composition of LNG in terms of major components?
   a) 80-90% methane, 5-10% ethane, 1-5% propane, and trace amounts of other gases
   b) 70-80% methane, 10-20% ethane, and 5-10% butane
   c) 50-60% methane, 20-30% ethane, and 10-20% propane
   d) 90-95% propane, 5-10% methane, and trace amounts of ethane

12) What is the primary principle behind the operation of bio electrochemical batteries (BEBs)?
   a) They generate electricity using only chemical reactions between metals.
   b) They use living microorganisms or enzymes to catalyze redox reactions and generate electrical current.
   c) They operate based on heat from the environment.
   d) They use radioactive materials to produce electrical energy.

13) Viscosity at 100°C of the sample oil (U) = 18 cSt
   Viscosity at 100°C of the standard oil with a low VI (L) = 12 cSt
   Viscosity at 100°C of the standard oil with a high VI (H) = 24 cSt
   Calculate the Viscosity Index (VI).
   a) 50 b) 60
   c) 56 d) 70

SECTION B

Q2 a) Give synthesis properties and applications of Thermosetting polymer **05**
b) Give properties and applications of CNT **05**

Q3 a) writes a note on electrochemical corrosion **05**
b) Explain corrosion testing methods **05**

Q4 a) Explain composition, properties, and applications of CNG **05**
b) Writes a note on alkaline fuel cell **05**

Q5 a) Explain synthesis properties and applications of PVA **05**
b) Calculate total hardness of water containing following impurities **05**

Impurity	Wt	Molecular wt.
CaCo3	13.5ppm	100
MgSO4	65.2ppm	120
Ca(HCO3)2	21ppm	162

Q6 a) write a note on electroplating **05**
b) Explain casting, forging, rolling, machining **05**

Q7 a) Define fuel give its classification and its unit **05**
b) Explain proximate analysis **05**